A Plot-Level Hazard and Livability Screen for Hill Land

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Abstract

National and state-level hazard maps are built to guide planning decisions at the scale of a district or a watershed. They have no demonstrated skill at the scale that matters to a buyer, lender, or developer: one plot of land. This paper describes a screening model that scores individual hill-land parcels in Uttarakhand and Himachal Pradesh, India, for landslide, flood, and glacial-lake-outburst hazard, and for livability, at 30-metre ground resolution inside each of 136,935 scored patches. The model is built from satellite elevation, open hazard and climate records, and road-network access data, scored against fixed, pre-declared thresholds. It is tested against landslide, flood, and outburst events the model had not seen, and separately against live property-listing prices, a test the model was not built to pass. Results, data sources, and stated limits follow.

1. Introduction

Hill land plots that look alike on a brochure behave very differently on the ground. Two parcels a few hundred metres apart can sit on stable terrain and on an active slide path respectively, differ by tens of metres in distance from a seasonal torrent, or fall on opposite sides of a glacial-lake-outburst corridor — none of which shows up in a listing photo or a plot boundary survey.

Hazard information that does exist is built for a different scale. National landslide susceptibility maps and state disaster atlases are produced to guide district- or watershed-level planning. Tested directly against documented landslide events recorded after the model was built, a national susceptibility layer of this kind showed no usable skill at the scale of an individual parcel. The gap is not a flaw in those maps — they were not built for plot-level decisions — but it leaves buyers, lenders, developers, and the lawyers and brokers who advise them without a number that applies to the specific plot in front of them.

This paper describes a model built to close that gap: one score, and a set of underlying measurements, per plot, each traceable to a named source, each tested against events the model did not see during development.

2. Coverage

The screen covers, starting with Uttarakhand and Himachal Pradesh:

- 136,935 scored patches, each approximately 0.8 km² — an H3 hexagonal cell at resolution 8.
- 2 states, 20 districts, 200 tehsils (sub-districts).
- 30-metre ground resolution for terrain measurements inside each patch.
- 80 or more measured values, plus 30 climate and hazard rows (each given a fixed 1–5 score), per plot report, organized across 9 report sections.

The report structure was cross-checked line by line against two published international sample report editions for hill land. All 72 line items of those editions are accounted for in this product: each has either a matching data source and licence, or an explicit stated reason it is not, or not yet, included.

3. Method

The score for each patch is built from four layers of measurement. The exact thresholds, factor weights, and internal rule structure used to combine them are not published (Section 7); what follows describes what is measured and how it is checked, not the rules themselves.

3.1 Terrain geometry

Slope, elevation, aspect, and drainage are derived from satellite elevation data at 30-metre resolution. A row/column duplication artifact was found in the raw elevation export during development — a resampling effect of the export process, not a feature of the terrain — and corrected before any terrain-derived layer was computed from it.

3.2 Hazard exposure

An independent landslide-susceptibility estimate, river and torrent proximity, and traced glacial-lake and glacial-lake-outburst-flood corridors are computed for every patch. Where two or more of these layers cover the same ground, they are cross-checked against each other.

3.3 Climate and access

Heat, frost, fire weather, drought, heavy rainfall, and wind are drawn from open climate records. Access is measured as road-network travel distance — not straight-line distance — to the district capital, the nearest highway, the nearest hospital, the nearest school, and the nearest market town.

3.4 Fixed, pre-declared thresholds

Every hazard band and flag is set against a physical cutoff decided before scoring begins. None of these cutoffs are rescaled relative to whatever else happens to be on the map on a given run. The practical consequence: the same ground reads the same way regardless of what else is nearby or what else has been scored that day.

The remainder of this paper — validation results, market and quality-control checks,
limitations and data sources — is available on request.
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events, each against matched ground nearby — the same state block, the same terrain class, the same elevation band, and in some comparisons the same road-distance band.

Table [illegible]: Matched-ground backtest results

Flag or band	Comparison	Lift over matched ground
Tsunami + river flood flag	National landslide catalogue	[illegible]
Tsunami + river flood flag	Documented flood incidents	[illegible]
Glacial-lake outburst corridor	[illegible] of [illegible] documented outburst events, vs. [illegible] base rate on matched ground	[illegible]

The observed maximum flood grade was checked for accuracy (AUC) against an independent satellite-derived flood map: [illegible] overall, and [illegible] for locations with more than one recorded flood.

[illegible] Event-fitted scores against [illegible] set-aside events

The combined hazard score is a model fitted to recorded events. Its landslide and flood terms were fitted (shallow gradient-boosted trees, fixed settings) on [illegible] recorded landslide, flood and outburst events with exact or village-level locations, using terrain, channels and height above them, rainfall, soil, vegetation and built-up ground; the flood term is averaged half and half with the physical flood score, and the earthquake term is unchanged. Another [illegible] events ([illegible]) were set aside before fitting, split by whole map cells so that neighbouring events stay on one side, and balanced by state block and event type. Every design choice was made on the [illegible] events, which were inspected over three rounds of changes; the set-aside events were scored once for the decision. A robustness check run afterwards (comparison ground inside the set-aside cells, which the model had not seen in training) gives the figures in Table [illegible]; with comparison ground the model had seen, the first read gave an AUC of [illegible]. Each set-aside event was compared with ground of the same type in the same region that had no record: same state block, terrain class, river setting and road-distance band, using comparison ground the model had not seen.

Table [illegible]: Event-fitted score against matched ground, [illegible] set-aside events

Measure	Result
Accuracy (AUC), [illegible] chance; [illegible] of the [illegible] events were had no comparison ground	[illegible] ([illegible] CI [illegible]) in [illegible] cells, resampled by map cell, [illegible] [illegible]

Gain over the outlier physical score	[illegible] (95% CI [illegible]), unchanged by map cell
Accuracy (AUC), outlier physical score on the same events	[illegible] (95% CI [illegible])
Events in the highest-ranked fifth of matched ground	[illegible] (20% expected by chance)
Events in the highest-ranked tenth	[illegible] (10% expected by chance)
Events in the lowest-ranked fifth	[illegible] (20% expected by chance)
Events on the top half of the land by score (50% of land)	[illegible]
Events in the lowest-scored [illegible] of land	[illegible]
Landslide score on landslide events / flood score on flood events	[illegible] / [illegible]
Western / eastern / Himachal block	[illegible] / [illegible] / [illegible]

Checks on this result: a random same-type comparison patch given to each event scores [illegible]; dropping any one map cell leaves the AUC between [illegible] and [illegible]; the gain holds for landslides and floods, for events before and after [illegible], and for news and government records. It is smallest for the Geological Survey of India mapped slides in Himachal (older inventory slides), where the change is not distinguishable from none. The outlier version (weights fixed from physical reasoning) scored [illegible] on [illegible] events and [illegible] on [illegible] set-aside events; the sets differ, and that outlier set-aside group held few western-block events ([illegible] of [illegible]), so the table's [illegible] is the like-for-like comparison. The share-of-land rows describe the same events and are not a separate test.

5. Market validation

A separate check was run against live market data rather than against disaster records. Asking prices were collected from several property portals (plots only). The largest portal was used as the guide: its listings were cleaned by robust outlier removal within each town and logic checks, with no fixed price range, and other portals' listings were kept only where they fell within a factor of four of that guide. Prices are expressed as price per square metre relative to the median of each town; listings that name a local land unit (biswa, bigha, kanal, marla, nali) were not priced. Prices were compared at the level of the locality, not the individual listing, since many listings in one locality share the same geocoded point. Each locality is scored as the mean of its hexagon and the six neighbours.

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- [illegible]

- [illegible]

- [illegible]

- [illegible]

[illegible]

6. Quality control

[illegible]

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- [illegible]

- [illegible]

were then independently reviewed for geographic plausibility. Zero confirmed anomalies in either group.

7. Limitations

- This is a relative-risk screening tool, built on a model tested against historical disaster records and compared with asking prices. It does not replace official certifications or the qualified engineering, legal, and administrative requirements a purchase, loan, or build still needs.
- The landslide event samples are modest in size ([illegible], [illegible], [illegible], [illegible] and [illegible] events respectively). The lift and accuracy figures in Section [illegible] should be read as direction, not as an exact probability for any one plot.
- The event-fitted score uses built-up ground, so part of its ranking reflects where events get recorded (settled ground and roads). On the [illegible] [illegible] events, matching comparison ground on built-up share kept more than half of the gain over the physical score, not all of it. The landslide term also takes a published susceptibility map as one input, and that map's training events may overlap the test events. The test events on map cells: [illegible] of the [illegible] events were on hill or high-mountain terrain and [illegible] on plains, so the event-fitted score is reported for sloping hill land only and the physical drivers stand alone on flat ground, plains and foothills. The Geological Survey of India mapped slides in Himachal [illegible] least. The earlier version (landslide driver without the susceptibility map [illegible] built-up share alone [illegible] without matching) describe the earlier physical score.
- Event records skew toward locations near roads and towns — a known reporting bias in disaster records generally.
- The market check in Section [illegible] uses asking prices, not sale prices. Over half of the localities have a single listing, and 99acres supplies many of the hill localities; without it the hill link is weaker ([illegible] interval includes zero). The score is a ranking, not a probability.
- The exact scoring thresholds, factor weights, and internal rule set are not published. Methods, full data sources, and complete test results are available on request.

8. Conclusion

The model scores [illegible] patches across Uttarakhand and Himachal Pradesh for hazard and livability, at [illegible] ground resolution, from measurements that trace to named open sources. It separates events it had not seen in development from similar nearby ground at rates above chance, as reported in Section [illegible], and its livability score shows a modest link with asking prices on hill land, mostly through 99acres, as reported in Section [illegible]. Both checks are reported with their sample sizes and confidence intervals, and the limits of each are stated in Section 7.

9. Data sources

- Copernicus DEM (elevation)
- OpenStreetMap contributors (ODbL) — road network and access
- ERA5-Land — climate records
- JRC Global Surface Water
- ICIMOD glacial lakes
- Randolph Glacier Inventory
- Geological Survey of India landslide inventory
- WorldPop